

Introduction to SOC

1. What is a SOC?

- **SOC (Security Operations Center)** is a **centralized team & place** where security professionals monitor, detect, analyze, and respond to cyber threats.
- Think of it as the **"control room" of cybersecurity** in an organization.
- Purpose:
 - Protect company data, systems, and networks from cyber-attacks.
 - Monitor 24/7 for suspicious activity.
 - Act quickly when a threat happens.

👉 **Analogy:** Just like a police control room monitors CCTV cameras in a city, the SOC monitors the company's digital environment.



2. Roles & Responsibilities of a SOC Analyst

SOC Analysts are usually divided into **tiers** (levels) based on expertise:

Tier 1: Alert Monitoring (First Responders)

- Monitor alerts from SIEM tools (like Splunk, QRadar).
- Triage alerts → check if they are **real threats** or **false alarms**.

- Document findings and escalate if needed.

👉 **Example:** Like a call center agent who takes the first call, checks seriousness, and forwards it to the right department.

Tier 2: Incident Responder (Deep Investigation)

- Investigate alerts escalated by Tier 1.
- Perform log analysis, malware analysis, and network traffic inspection.
- Contain and mitigate threats (e.g., isolate infected machine).
- Provide reports on incidents.

👉 **Example:** Like detectives who investigate deeper after police patrol reports an issue.

Tier 3: Threat Hunter / Subject Matter Expert

- Handle **advanced and complex attacks** (APT, zero-day exploits).
- Proactively hunt for hidden threats (not just responding to alerts).
- Provide threat intelligence and tuning for SOC tools.
- Create playbooks and train Tier 1 & 2.

👉 **Example:** Like cyber crime experts who research criminals' techniques and prevent future attacks.

SOC Workflow and Escalation Process

SOC works in a structured way to handle security events:

1. **Monitoring:** SIEM tools generate alerts.
2. **Detection:** Tier 1 analyst checks if alert is real or false.
3. **Escalation:** If real → escalate to Tier 2 for deeper analysis.
4. **Investigation:** Tier 2 investigates scope, impact, and root cause.
5. **Response:** Contain and remove the threat.
6. **Recovery:** Systems are restored, patched, or cleaned.
7. **Reporting & Lessons Learned:** Documentation + improving defenses.

👉 **Key Point:** Escalation ensures that serious threats reach more skilled analysts quickly.